

Ionic Liquid-Modified Porous Materials for Gas Separation and Heterogeneous Catalysis

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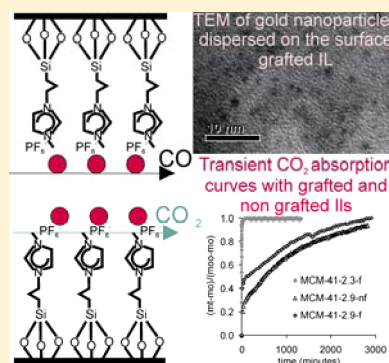
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S Supporting Information

ABSTRACT: This work examines important physicochemical and thermophysical properties of ultrathin ionic liquid (IL) layers under confinement into the pore structure of siliceous supports and brings significant advances toward understanding the effects of these properties on the gas separation and catalytic performance of the developed supported ionic liquid phase (SILP) and solid catalysts with ionic liquid layers (SCILL). SILPs were developed by making use of functionalized and nonfunctionalized ILs, such as 1-(silylpropyl)-3-methyl-imidazolium hexafluorophosphate and 1-butyl-3-methyl-imidazolium hexafluorophosphate ILs, whereas the SCILL was prepared by effectively dispersing gold nanoparticles (AuNPs) onto the IL layers inside the open pores of the SILP. The information derived from the gas absorption/diffusivity and heterogeneous catalysis experiments was exemplified in relation to the liquid crystalline ordering and orientation of the IL molecules, investigated by X-ray diffraction (XRD) and modulated differential scanning calorimetry (MDSC). The extent of pore blocking was elucidated with small angle neutron scattering (SANS) and was proven to be a decisive factor for the gas separation efficiency of the SILPs. CO₂/CO separation values above 50 were obtained in cases where liquid crystalline ordering of the IL layers and extended pore blocking had occurred. The presence of the IL layer in the developed SCILL assisted the formation of ultrasmall (2–3 nm) and well-stabilized AuNPs. The low-temperature CO oxidation efficiency was 22%. The catalytic experiments showed an additional functionality of the IL, acting as an “in-situ trap” that abstracts the product (CO₂) from the reaction site and improves yield.



1. INTRODUCTION

High specific surface area materials making use of ultrathin ionic liquid (IL) layers as surface functionalities constitute very promising systems for a variety of applications including heterogeneous catalysis and gas separation.^{1,2} To this end, important advances can arise with the development of supported ionic liquid phase (SILPs) systems and solid catalysts with IL layers (SCILLs), owing to the low volatility, exceptional solvation and miscibility, thermal stability, and tunable physicochemical properties of ILs that can be tailored according to the requirements of a specific application. Notably, due to the fact that ILs are expensive, their fine deposition on a supporting surface would also be necessary to render feasible their application at a large scale.

Several groups apply delicate in situ techniques to develop and further investigate SILPs at the microscopic level.^{3,4} Physical vapor deposition (PVD) under ultrahigh vacuum (UHV) conditions is an emerging methodology involved in preparing precisely controlled and high surface purity ultrathin IL layers and studying the elementary chemical and physical processes at the IL interfaces (IL-gas, IL-liquid, IL-solid), as well as layering, orientation and ordering phenomena, and initial growth

behavior. All these aspects may significantly control the chemistry of IL-based reaction and separation systems, and the PVD approach offers a broad range of new experimental possibilities in this direction. IL films for surface science investigations may be prepared also by ex situ methods. Thick layers can be formed by either dipping the support into the IL or by mechanically spreading the IL on the support surface. Thinner more homogeneous layers may be prepared by spin coating on a surface.

A common feature of these techniques is the use of a metal or metal oxide planar surface as IL support. Atomically flat and well ordered alumina films can be grown by oxidation and annealing of NiAl(110).⁵ These films, the structure of which has been analyzed at the atomic level,^{6,7} have been used in a large number of model studies in heterogeneous catalysis over the recent years.^{8–11} As the film is thin enough to avoid charging, it is ideally suited for photoelectron and scanning tunneling microscopy

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studies. Also, IR studies in grazing reflection geometry can be performed, similar to metal surfaces.^{12–14}

The remarkable potential of the “surface science and model catalysis” type of approach to provide a deep understanding of SILPs at the microscopic level must be further related and complemented with the study of “application-oriented” SILPs to make possible the elucidation and establishment of valid correlations between the ILs interface properties and the absorption/separation or catalytic performances of SILPs and SCILLs. For instance, ex situ dipping techniques seem more feasible at the upscaled level, and moreover, the use of atomically flat and well-ordered surfaces is far from the real case of highly porous supports. In fact, it is useless to exploit solely the external surface of a material when most of its high specific surface resides into its pore structure. To add to this, PVD techniques do not ensure the deposition of IL layers on the entire pore surface.

In addition to the aspects related to the interaction of the IL with the pore surface and the characteristics of the IL–gas interface, there are other major important issues that demand rigorous investigations when IL deposition is attempted in porous supports. The lack of deposition control when applying ex situ dipping techniques necessitates the existence of analytical methods that will allow a deep insight on the extent of pore blockage, a very important parameter for heterogeneous catalytic and gas separation applications. For example, when the IL is used to catalytically immobilize active transition metal complexes, it will be inefficient to completely block the pores and force the gas reactants to diffuse slowly through thick and viscous IL layers in order to reach the reactive sites. Additionally, in SILP systems developed for gas separation, the remaining open pore structure may considerably degrade the overall separation performance due to the fact that the gas solvation properties of the applied IL may be counterbalanced by the excess sorption of the gaseous molecules in the void pore volume.

In this work we have applied ex situ dipping techniques to develop SILPs that consist of an imidazolium-based IL (1-butyl-3-methyl-imidazolium hexafluorophosphate-[bmim][PF₆[−]]) in functionalized (1-(silylpropyl)-3-methyl-imidazolium hexafluorophosphate [spmim][PF₆[−]]) or nonfunctionalized form, on ordered (MCM-41, SBA-15) or nonordered (Vycor, Controlled Pore Glass-(CPG)) mesoporous silica supports. Ordered mesoporous silica modified with a silylated-IL was further utilized to finely disperse gold nanoparticles (AuNPs), where the IL acts both as a stabilizer and as an absorbent that rapidly abstracts the reaction products from the active metal catalytic sites, thus, enhancing reactivity. SILPs and AuNPs-functionalized SILPs were studied in regard to their gas separation performance and catalytic efficiency for low temperature CO oxidation, respectively.

In addition, samples of ordered mesoporous silica of various pore diameters ranging from 2.3 to 5.6 nm were studied with the purpose to facilitate the introduction of the IL molecules into the non tortuous support channels and investigate the effect of pore dimension and confinement on to the liquid crystalline ordering of the supported phase. The high amount of IL loading allowed us to monitor the liquid crystalline ordering with methods such as modulated differential scanning calorimetry (DSC) and X-ray diffraction (XRD), whereas comparison between ordered and nonordered pore structures of similar pore size revealed that network tortuosity did not affect the deposited amount of IL. Small-angle neutron scattering (SANS) was used to elucidate the extent of pore coverage, and exemplify issues regarding misleading conclusions obtained by the typically applied, yet

pore-mouth sensitive, techniques such as liquid nitrogen (LN₂, 77 K) porosimetry. Finally, the performance characteristics of the developed materials for gas diffusion/absorption/separation of CO₂/CO and low temperature oxidation of CO are discussed in regard to the possible orientation of the deposited IL molecules, taking into consideration recent reports on the quantitative analysis of XP spectra,¹⁵ a relatively straightforward technique that provides information on the molecular orientation, surface enrichment, and ion layering at the IL/vacuum or the solid/IL interface.

2. EXPERIMENTAL SECTION

2.1. Preparation of the SILPs Using the Functionalized

IL. A total of 200 mg of the porous support was dried at 423 K in vacuum for 1 h. A solution of triethoxy chloropropyl silane (2.5 mL, 10.38 mmol) in 10 mL of dry CHCl₃ was added to the dried powder, and the mixture was refluxed under a N₂ atmosphere for 24 h. The solution was then filtered, and the remaining solid was washed sequentially with pentane (30 mL) and acetonitrile (30 mL) and refluxed with diethyl ether (30 mL). After filtration, the material was dried at 343 K in vacuum for 1 h. Then 1-methyl imidazole (5 mL) was added and the mixture was rotated at 358 K under a N₂ atmosphere for 48 h. The mixture was filtrated, and the remaining powder was refluxed with diethyl ether (30 mL) for 30 min. After filtration the powder was dried again at 343 K under vacuum for 1 h. Subsequently, a solution of NaPF₆ (68 mg, 0.4 mmol) in EtOH (6 mL) was added to 64 mg of the dried powder, and the mixture was rotated for 5 days under a N₂ atmosphere. AgNO₃ test was positive from the first 24 h of reaction onset. The reaction mixture was filtered and the sample was washed by reflux with MeOH (30 mL) for 15 min. After filtration, the sample was dried in vacuum for 1 h.

2.2. Preparation of the SILPs Using the Nonfunction-

alized IL. For the preparation of the SILP, a special glass device was constructed that allowed the regeneration of up to 150 mg of the porous support in high vacuum (10^{−5} mbar) and temperatures up to 180 °C. Upon completion of the outgassing (1 day), the temperature was decreased under vacuum. At room temperature, the vacuum valve was closed and the support was soaked with a certain amount (10 mL) of [bmim][PF₆[−]] solution (0.2 M in MeOH) that was introduced with a syringe through a septum, which was appropriately mounted to the device. Details of the synthesis of the nonfunctionalized IL 1-butyl-3-methylimidazolium hexafluorophosphate ([bmim][PF₆[−]]) are provided in the Supporting Information.

2.3. Preparation of the SCILL. A total of 40 mg of the SILP material [spmim][PF₆[−]]/MCM-41–2.9 nm (MCM-41–2.9-f) was dried at 100 °C under vacuum for 1 h. Upon completion of the outgassing, AuCl₃ (0.52 mmol) in 1.5 mL of dry DMF was added, and the mixture was left under inert conditions and rotation for 24 h at room temperature. Then the mixture was heated to reflux for 1 h and subsequently centrifuged at 7000 rounds per minute in order to obtain the Au-functionalized hybrid material (SCILL). The material was further washed with 5 mL of ethanol and centrifuged at 7000 rounds per minute to obtain the final product. Details of the extraction procedure applied for washing out externally deposited AuNP aggregates are provided in the Supporting Information.

2.4. Analytical Instruments. XRD analysis of the SILPs was carried out in a D8 Advance Bruker diffractometer using Cu K α radiation and parallel beam stemming from a Göbel mirror. The measurements were performed in a 2 θ region between 15 and 55°. XRD of the SCILL was carried out in a Siemens D500 X-ray

diffractometer. Transmission electron microscopy was performed with a FEI CM20 TEM at 200 kV. Modulated DSC analysis was performed using a 2920 MDSC (TA Instruments). Gas adsorption characterization was performed on an IGA gravimetric analyzer (IGA, Hiden Analytical), and LN_2 porosimetry was performed with an Autosorb-1 MP porosimeter (Quantachrome).

3. RESULTS AND DISCUSSION

3.1. Amount of IL Deposited in a Porous System by Ex Situ Dipping. Factors that affect the amount of IL loaded into the channels of a porous support are studied on the basis of LN_2 porosimetry results (Table 1). These are the total pore volume

Table 1. Surface Area, Pore Size, and Volume of the Samples Before and After the Modification with the ILs

	BET ($\text{m}^2 \text{g}^{-1}$)	pore size (nm)	total pore volume (mL g^{-1})
Vycor	175	4.3	0.24
CPG	153	20.1	0.85
SBA-15	711	5.6	0.93
MCM-41-3.3	1117	3.3	1.54
MCM-41-2.9	1068	2.9	1.18
MCM-41-2.3	1543	2.3	0.86
Vycor-f ^a	96		0.12
CPG-f	147		0.71
CPG-nf ^b	96		0.61
SBA-15-f	359		0.65
SBA-15-nf	14		0.05
MCM-41-3.3-f	430		0.84
MCM-41-2.9-f	368		0.42
MCM-41-2.9-nf	113		0.33
MCM-41-2.3-f	41		0.08

^aA functionalized IL is abbreviated with -f. ^bA nonfunctionalized IL is abbreviated with -nf.

and surface, the surface chemistry (functional groups) and charge, as well as morphological characteristics of the pore structure such as the dimension of the pores, the pore size distribution, and the tortuosity of the network. The volume of IL loaded during ex situ dipping using the functionalized and nonfunctionalized IL was calculated from the pore volume difference between the pristine support and the SILP system.

When a silylated IL was applied as surface modifier, there was strong correlation between the BET surface area of the pristine support and the pore volume reduction after dipping, the later expressed as volume of IL loading (Figure 1a). No correlation was observed between the IL loading and the initial pore volume or pore dimension (Figure 1b). In this regard, the formation of a monolayer of the functionalized IL on to the pore walls of the support can be confirmed. Small deviations from linearity (Figure 1a) can be attributed to the different population and proximity of the silanol ($\text{Si}-\text{OH}$) groups on the surface of the different supports. In general, the silanol density of ordered mesoporous silicas (e.g., MCM-41 or SBA-15) ranges between 1 and 3 SiOH/nm^2 ,^{16–18} whereas for porous glasses like Vycor and CPG, silanol densities between 1.6 and 3.8 SiOH/nm^2 have been reported.¹⁹ However, the population of surface silanols and their distribution on the mesopore surface strongly depends on thermal treatment, activation under vacuum, and dehydroxylation/rehydroxylation processes.

For the case of a standard SBA-15, a silanol density closer to 1 SiOH/nm^2 has been observed when the material is heated at 200 °C under vacuum,¹⁸ conditions which were similar to these applied in the current work to regenerate the samples before the LN_2 measurement and the ex situ dipping process. On the other hand, in a recent study of our group,²⁰ a water vapor desorption method was developed and applied that allowed for the definition of the surface silanol population. By this method, silanol densities of 1.3 and 2 SiOH/nm^2 were obtained for Vycor and MCM-41, respectively. In Figures 1c it is shown that plotting

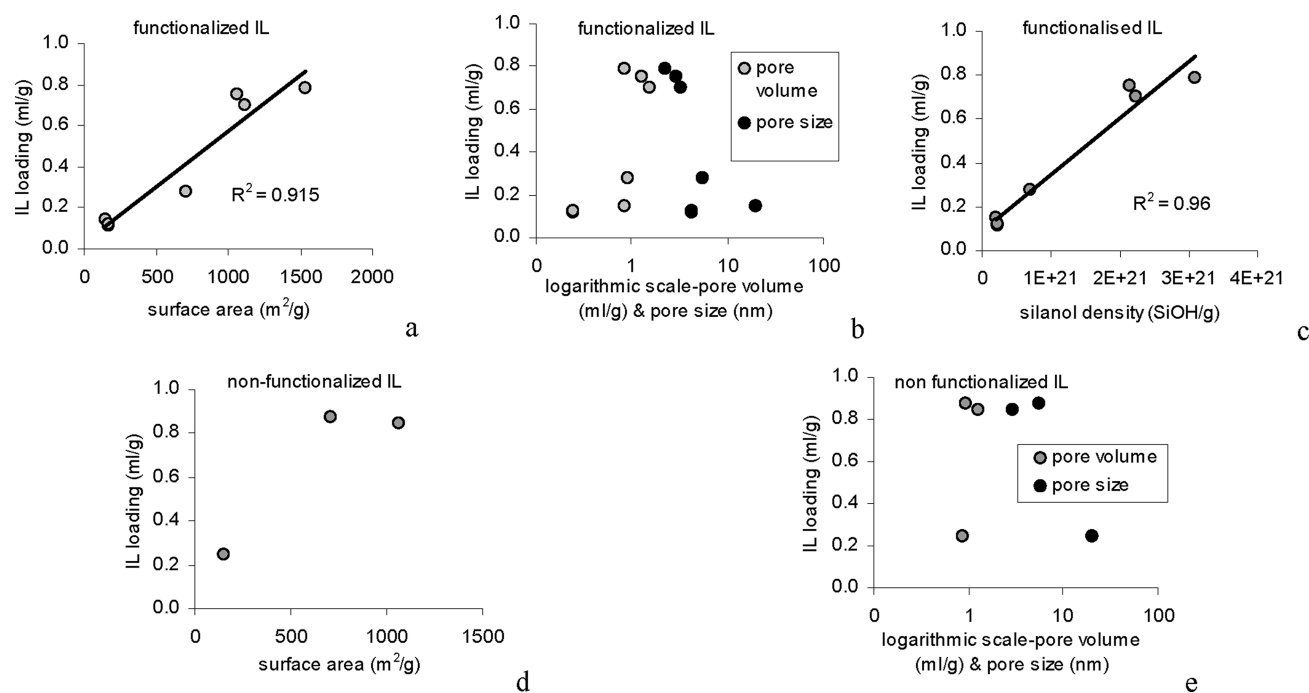


Figure 1. Correlation of the IL loading with (a, d) the surface area of the supports, (b, e) the pore volume and pore size, and (c) the surface silanol density.

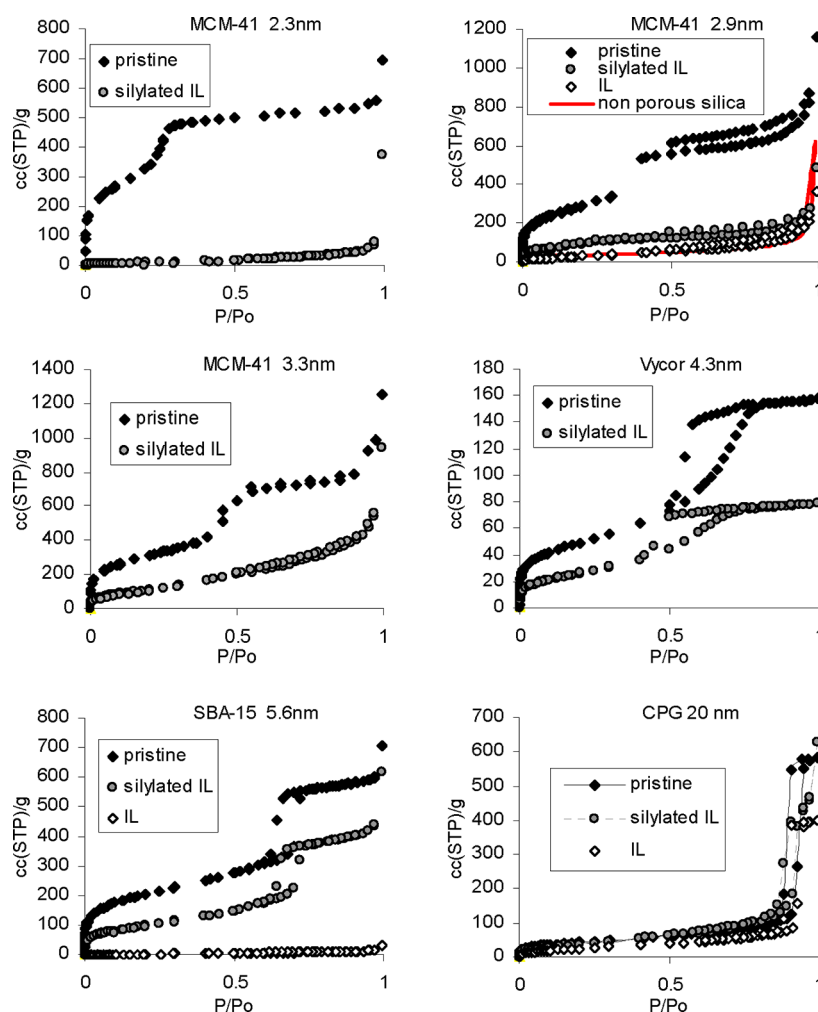


Figure 2. LN_2 porosimetry of the porous supports and the relevant SILPs.

the IL loading versus the experimentally defined surface silanol population of the solid supports readily improves the correlation factor from 0.92 (IL loading vs surface area) to 0.96. Deviation from the linearity remains solely for the MCM-41 samples of different pore size. The reason behind this is that MCM-41 is characterized of at least three types of silanol groups: single silanols $(\text{SiO})_3\text{SiOH}$, hydrogen-bonded $(\text{SiO})_3\text{SiOH}\cdots\text{OHSi}(\text{SiO})_3$, and geminal silanols $(\text{SiO})_2\text{Si}(\text{OH})_2$ with different accessibility to silylating agents and different relative population among samples. It is also worth noting that there is no deviation from the linearity between the tortuous (Vycor, CPG) and nontortuous samples (MCM-41, SBA-15), a fact indicating that the interconnectivity and constrictions of the channel network do not affect the proper wetting of the pore walls with the IL solution during the ex situ dipping process.

The absence of a good correlation between the amount of IL loaded and any of the pore structure characteristics for the supports modified with the nonfunctionalized IL is demonstrated in Figure 1d,e. The main reason is that the interactions of the IL with the pore surface are limited to dipole interactions (there is not chemical bonding) and, consequently, a random leaching of the deposited IL layers during the sequential washings with the solvent might have occurred. Even though interactions with the surface are weak, we note that the amount of IL loading was higher for the cases when the nonfunctionalized $[\text{bmim}][\text{PF}_6^-]$ was applied as support modifier (Table 1). The

smaller molecular volume compared to the silylated IL, studied also under different deposition orientations, with both anions and cations in direct contact with the surface may generate enough intra pore space for the formation of succeeding layers, thus, enhancing the loading. Extended discussion on the possible IL molecular orientation in relation to the obtained gas diffusivity data and recent findings by other groups that apply techniques such as photoelectron and surface vibrational spectroscopy, is given in sections 3.3 and 3.4. In the next section, we discuss the weakness of LN_2 to provide information on the extent of pore blocking by the deposited IL layers, and we highlight the potential of SANS as a pore bulk sensitive method to provide an insight on this issue.

3.2. Extent of Pore Coverage. Complete pore coverage or blocking by the deposited IL phase may be detrimental for a catalytic process where the high pore surface must be exploited to the maximum. On the contrary, this may be a requirement for a gas separation application, especially when a task-specific IL (e.g., amino-functionalized for CO_2 sorption) is involved as surface modifier. The results of LN_2 porosimetry (Figure 2) alone cannot lead to safe conclusions about the extent of pore coverage mainly due to the following two reasons: (1) LN_2 porosimetry is a pore mouth-sensitive technique and, for example, the almost zero adsorption capacity observed for the MCM-41 (2.3 nm) modified with the silylated IL (Figure 2a) and for the SBA-15 (5.6 nm) modified with the nonfunctionalized IL (Figure 2e)

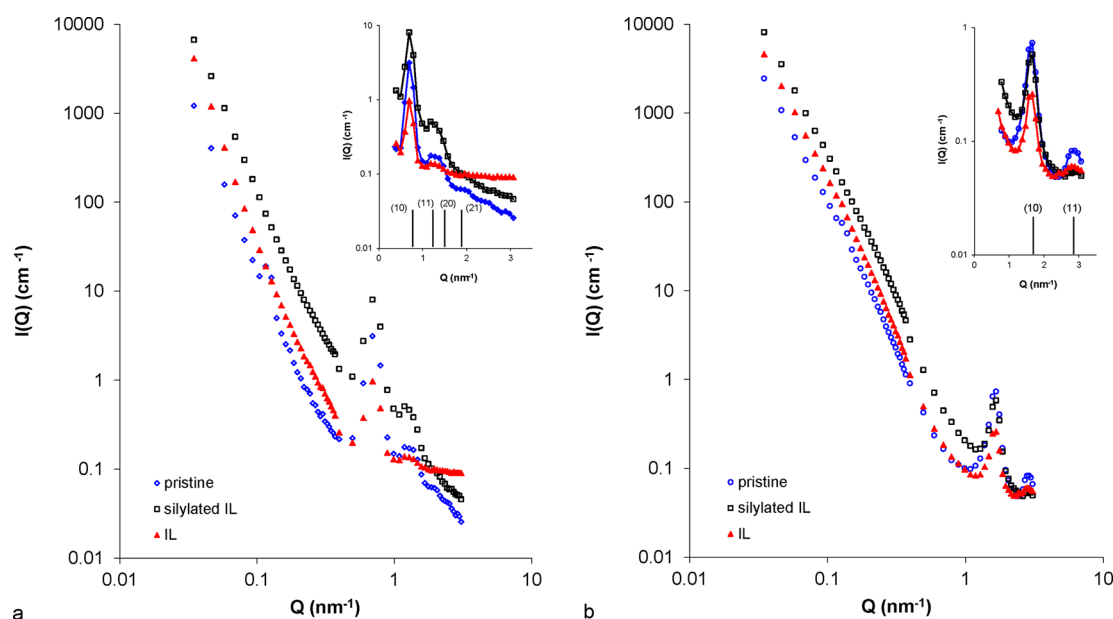


Figure 3. SANS curves from pristine, silylated IL, and IL loaded in (a) SBA-15 (5.6 nm) and (b) MCM-41 (2.9 nm); insets: high- Q region spectral details; lines have been added as a guide to the eye for the peak positions; peak positions are also indicated by vertical lines.

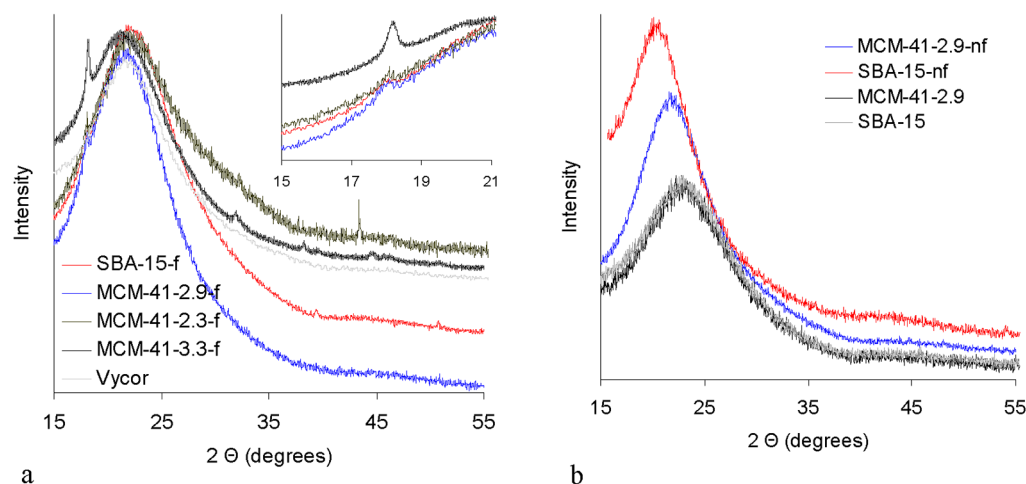


Figure 4. (a) Liquid crystalline ordering by XRD was concluded solely for the SILPs developed with the functionalized IL; (b) XRD spectra of the supports and relevant SILPs developed with the nonfunctionalized IL.

might be the result of pore blocking due to accumulation of the deposited IL close to the pore entrance. Consequently, a zero N_2 adsorption capacity does not mean complete coverage of the pore surface by the IL. (2) In some cases there is uncertainty as to whether the observed N_2 uptake is the result of adsorption in the interparticle voids, the pore voids, or both.

Besides the cases of the Vycor modified with the silylated IL, the CPG modified with both kinds of IL, and the SBA-15 modified with the silylated IL, where it is evident that the pore cores have remained open, for all the other samples it is not safe to discriminate between the contribution of open pores and interparticle voids. The above statement is supported by the LN_2 adsorption curve of a nonporous fumed silica (red line in Figure 2b) that converges with this of the MCM-41 (2.9 nm) sample modified with the nonfunctionalized IL.

To this extent, SANS can be a powerful technique for investigating whether the IL has been partially or completely filled within the pores of mesoporous materials. This is one of the advantages of the SANS technique over the conventional gas

adsorption methods because neutrons can also “see” pores that are inaccessible to gas molecules (such as closed pores). For instance, as mentioned previously, IL could have been dispersed around the pore entrance, thus, preventing the access to N_2 molecules.

Figure 3 shows the scattering curves from the pristine, silylated IL, and IL-loaded SBA-15 (5.6 nm) and MCM-41 (2.9 nm), respectively. In the case of MCM-41, the two-dimensional hexagonal arrangement of cylindrical mesopores produces one intense peak (10) and one weak peak (11), while the high-order peaks are not observed because they fall outside of the experimental Q window (Figure 3b). In the case of the pristine SBA-15, however, except for the strong (10) reflection, the overlapping (11), (20), and the faint (21) Bragg peaks are visible (Figure 3a). The scattering from SBA-15 sample loaded with IL becomes flat for $Q > 1.6 \text{ nm}^{-1}$, mainly due to the incoherent background arising from the hydrogen contained in the IL; this results in very weak (11) and (20) reflections and the absence of the (21) peak (Figure 3a). The presence of the incoherent

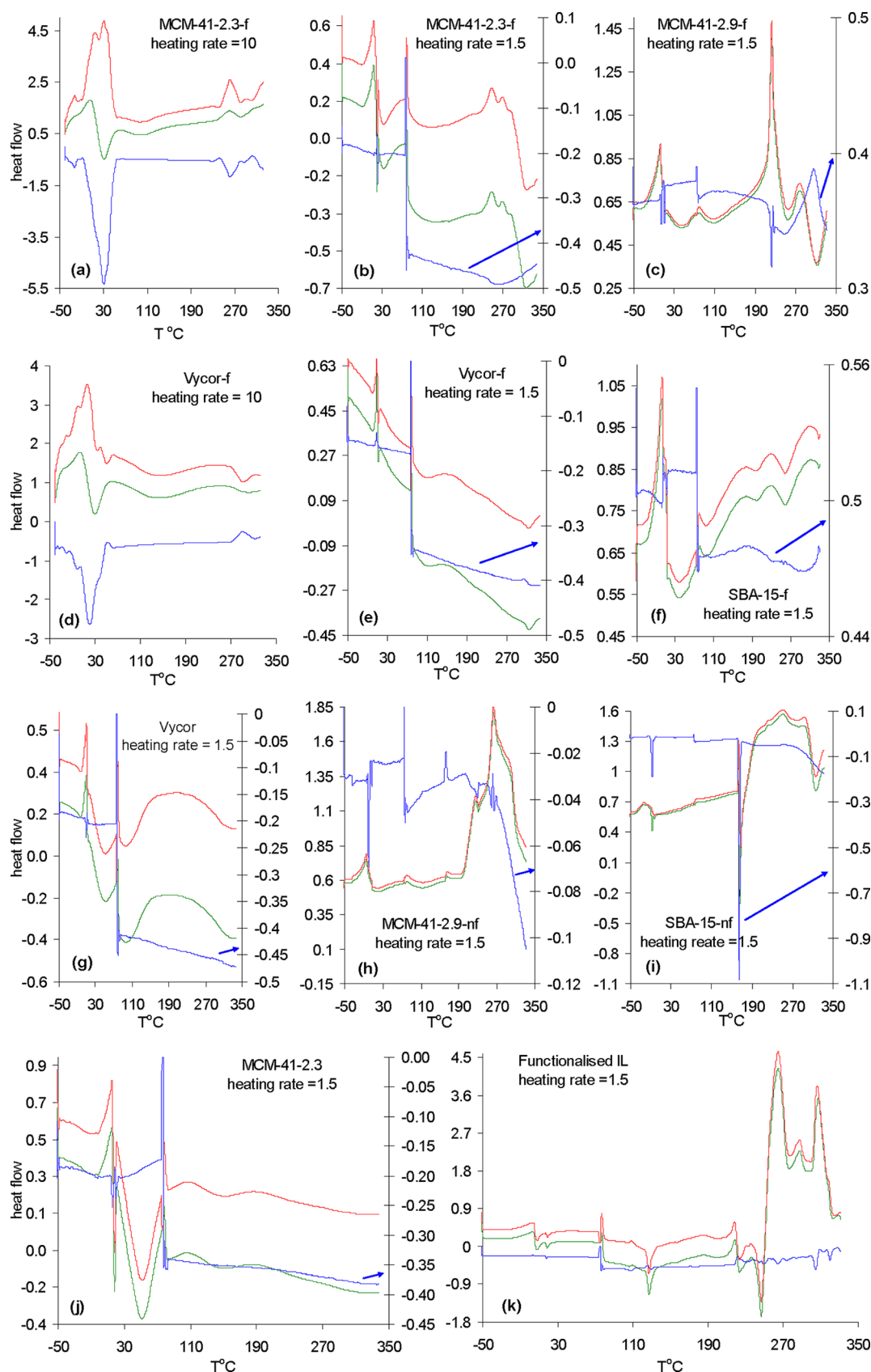


Figure 5. Modulated DSC thermographs of the developed samples. Blue lines correspond to the reversing component, red lines correspond to the nonreversing, and green lines correspond to the total (deconvoluted) thermograph. The heating rates and sample names are included in the plots.

background suggests that a larger amount of IL is confined on the pores compared to the amount of functionalized IL, in agreement

with the LN_2 results (Table 1). The position of the Bragg reflections and, thus, the lattice parameter for pristine and IL-

treated samples remains unchanged, implying a rather rigid solid matrix that does not change upon IL treatment. The low- Q range of the SANS signal follows a power law, arising from the intergrain interfacial scattering.²¹

According to the scattering theory, for a two-phase system (for instance, silica-empty pore and silica-pore completely filled with IL), the intensities of the Bragg reflections are related to the square of the contrast, defined as the difference of the scattering length density between the silica matrix and the pore content.^{22–24} In particular, from the calculated integrated ratio of the Bragg reflections of the pristine to the IL-loaded samples, one may conclude whether the pores are completely or partially filled with IL. The results suggest a complete and an almost complete pore filling with IL in the case of the IL loaded SBA-15 and MCM-41 samples, respectively, in agreement with nitrogen adsorption isotherms at 77 K (see also Table 1). On the other hand, the estimation of the skeleton density and, thus, the scattering length density of silylated-IL is not possible because of the presence of the silane part. However, the increased intensity of Bragg peaks for the silylated-IL SBA-15 compared to the silylated-IL MCM-41, suggests a smaller IL loading, also in agreement with LN_2 porosimetry results (Table 1). In particular, this intensity increase, especially of the first-order (10) reflection, has also been observed by the performance of in situ SANS^{25,26} and SAXS^{27,28} experiments during the initial stages of fluid adsorption on ordered mesoporous MCM-41 and SBA-15, respectively.

3.3. Structure Formation in IL Layers Confined into Pores. Due to the high loading of the IL phase, it was possible to observe X-ray diffraction reflections arising from the grafted IL phase. In all the XRD spectra (Figure 4), the broad reflection at about 22° is due to silica. Besides the case of the low surface area Vycor support where the IL loading was moderate, all the other SILPs that were developed using the functionalized IL as surface modifier (Figure 4a) exhibited a diffraction peak at around 18.2° , a feature suggesting liquid crystalline (LC) ordering of the silylated IL molecules in the grafted layer. The MCM-41–3.3-f sample presented the higher degree of ordering, as indicated by its intense Bragg peak at 18.2° and the additional diffraction peaks appearing at $2\theta = 31.9^\circ, 38.2^\circ, 39.4^\circ, 44.5^\circ, 46.2^\circ$, and 50.9° , which are attributed to the different orientation planes of the silylated IL inside the pores. In addition, sample MCM-41–2.3-f exhibits only the first two reflections, suggesting less ordering.

Comparing the samples developed on the several MCM-41 and SBA-15 supports, there is no evidence of a correlation between the intensity and number of their Bragg peaks with the pore structural characteristics such as the pore surface, size, and volume. In this regard, we can conclude that LC ordering of the functionalized IL molecules in the grafted monolayer is a function of the density of the surface silanol groups, which, however, may differ even between portions of the same sample depending on the thermal treatment and regeneration conditions. Notably, Bragg peaks were not observed for the SILPs loaded with the nonfunctionalized IL (Figure 4b).

Figure 5 presents the results of modulated differential scanning calorimetry (MDSC) for the SILPs prepared with the functionalized and the nonfunctionalized IL as well as the thermographs obtained for the pristine supports and the bulk functionalized IL ((1-(silylpropyl)-3-methyl-imidazolium hexafluorophosphate [spmim][PF₆[−]])). MDSC is a valuable technique to discriminate between reversible and nonreversible characteristics of thermal events, such as melting and decomposition, especially when these

occur at very similar temperatures. Focusing on the reversing component (blue lines) of the MDSC thermographs, it can be seen that there is an endotherm at about $260\text{--}270^\circ\text{C}$ that occurred only in the case of the SILP samples prepared with the functionalized IL (Figure 5a–c and f). When Figure 5a and b are compared, it can be observed that this endotherm was more pronounced with the faster heating rate ($10^\circ\text{C}/\text{min}$), a feature valid for all heating rate-dependent reversing transitions. This endotherm corresponds to the melting of the ordered IL phase, which was grafted onto the pore surface of the solid supports and appeared in all the samples that exhibited diffraction reflections in their XRD spectra (Figure 4a). As an exception, the endotherm was not observed for the Vycor sample modified with the functionalized IL because of the low amount of IL loading (see Table 1). The second endotherm that appeared solely in the nonreversing component (red lines) of the MDSC for all the supported IL samples at about 320°C corresponds to the decomposition of the confined IL phase. It is worthy to note that the decomposition temperature in the SILPs is lower than this reported for the bulk ionic liquid [bmim][PF₆[−]] (1-butyl-3-methyl imidazolium hexafluorophosphate), 350°C ,²⁹ and this is attributed to the presence of the highly acidic SiO₂ surface that acts as a catalyst.³⁰ Moreover, the sample SBA-15-nf, modified with the nonfunctionalized IL, presents an endotherm at the temperature of 150°C (Figure 5 i), which as referred in the literature,^{29,31} is attributed to the hydrolysis of the PF₆[−] anion in the presence of a trace amount of bind water that was not possible to be removed during the pretreatment at 120°C for 1 h. This issue raises questions concerning the applicability of the developed materials. However, as concluded by the thermographs, hexafluorophosphate hydrolysis occurred only in one of our SILP samples and solely for the case when the non-functionalized IL was used as the pore modifying agent. Finally, the reversing component of the thermograph for the bulk functionalized IL [spmim][PF₆[−]] does not comprise any thermal event at the low and high temperature area, showing that the cooling down to -50°C is not adequate for crystallization to occur. In this respect, the silylated IL exhibits supercooling phenomena, as is frequently observed with most of the ILs.^{32,33} Moreover, upon heating, the PF₆[−] anion hydrolysis endotherm appears again at the nonreversing component of the MDSC at a temperature of 130°C .

3.4. Orientation Aspects. Oriented layering of ILs on highly ordered surfaces is a characteristic investigated by many researchers.¹³ In our case, the silanization of the cation has a significant contribution and facilitates the ordered deposition of the synthesized IL phase. What is more important here is our discussion on the oriented layering of the deposited IL phase in relation to the CO₂ diffusivity at 35°C . For this study, we selected solely the SILP samples which, as evidenced by LN_2 porosimetry and SANS, underwent complete pore blockage upon IL deposition. The main idea was to exclude the contribution of open porosity on the overall diffusion mechanism and to focus on the characteristics of the IL phase.

We have selected CO₂ at the probe gas molecule because the system [bmim][PF₆[−]]/CO₂ is the best studied one, and the mechanism of CO₂ absorption and diffusion in the bulk IL phase has already been elucidated by many research groups through absorption, molecular simulation, and spectroscopic studies.^{34–36} Simulations indicate that CO₂ organizes strongly about the [PF₆[−]] anion in a “tangent-like” configuration that maximizes favorable interactions but is more diffusely distributed about the imidazolium ring. Kazarian and co-workers³⁴

investigated mixtures of CO₂ with ([bmim][PF₆⁻]) and ([bmim][BF₄⁻]) using ATR-IR spectroscopy. They found evidence of a weak Lewis acid–base interaction between the CO₂ and the anions of the ILs.

In the case of a SILP system, CO₂ dissolves in the immobilized-into-the-pores IL through the IL/gas interface close to the mouth of the pores and becomes ionized through its interaction with the anions. The ionized species diffuse along the length of the pore by successive jumps between the anions of the deposited IL. As shown in the schematic of Figure 6, a possible orientation of the

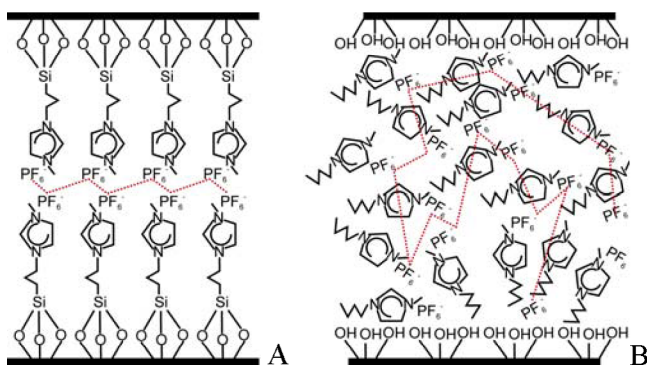


Figure 6. (A) Orientation of the silylated IL under confinement into the pores; (B) Random intrapore accommodation of a nonfunctionalized IL.

deposited IL phase with the anions toward the core of the pore (case A) may considerably enhance diffusivity when compared to a completely random orientation (case B) that leads to a random walk of the diffusing species between the anions. The use of a functionalized (silylated) IL, in our case, 1-(3-silylpropyl)-3-methylimidazolium hexafluorophosphate, which can be grafted on the surface of the support through covalent bonding between the alkoxy groups and the surface silanol groups, certainly enhances the possibility of a deposition orientation like the one presented in Figure 6A.

The transient curves of CO₂ adsorption of two successive pressure steps (0–250 mbar, 250–500 mbar) presented in Figure 7 verify the proposed orientation. By fitting the kinetic results with the appropriate solution of the transient diffusion equation for spherical particles³⁷ it was possible to derive the diffusivity constants (D/r^2 (1/min) where r = radius of sphere). The values for the highly ordered samples MCM-41–2.3-f and MCM-41–3.3-f were from 100 to 600 times higher than these of the less-ordered ones, like MCM-41–2.9-f, and of the non-ordered at all, such as the MCM-41–2.9-nf and SBA-15-nf. It is

interesting to note that our conclusion, correlating the high CO₂ diffusivity with the degree of the IL layer ordering, holds also between the two most crystallized SILPs. Indeed, the MCM-41–3.3-f sample, presenting an intense Bragg peak at 18.2° and a plurality of reflections of significant intensity (Figure 4a), exhibited a diffusivity constant ($D/r^2 = 0.05 \text{ min}^{-1}$) twice that of the MCM-41–2.3-f ($D/r^2 = 0.02 \text{ min}^{-1}$).

3.5. Effects of Loading AuNPs on a SILP. We next proceeded with the development of a solid catalyst with IL layers (SCILL) by further modifying the MCM-41–2.9-f sample in a DMF solution of AuCl₃ salt. Metal ion reduction in the presence of stabilizing agents, such as surfactants or polymers, has been used for fabrication and control of metallic nanoparticles.³⁸ The motivation to synthesize gold-nanoparticle-based SCILLs comes from the potential of gold nanoparticles to catalyze a variety of reactions at relatively low temperatures, in particular, oxidation of CO⁴⁰ and the twin role that the IL can play as particle stabilizer and preferential sorbent for CO₂, targeting a high-yield hybrid sorption/catalytic system. The as-produced SCILL sample underwent several washings in ethanol and subsequent centrifugations in order to remove the excess DMF, whereas, for a portion of the washed sample, a further extraction treatment (see Supporting Information) was applied to possibly wash out larger AuNPs that may have been deposited on the external surface of the SCILL (outside its pore structure). Indeed, from a comparison between the respective LN₂ isotherms of the SCILLs (Figure 8a), we can conclude that the extraction procedure led to the unblocking of a significant fraction of the pore structure. Moreover, after extraction, the SCILLs' pore volume remained lower than this of the respective SILP, a fact indicating the stability of the deposited nanoparticles and IL phase.

In regard to the gas absorption properties of the developed SCILLs, we expected that the deposition of AuNPs onto the pore-grafted IL should not alter the CO₂ absorption capacity and conclude to the significant enhancement of the CO uptake of the respective SILP. This was the case for the SCILL sample after the extraction of gold (Figure 8b). On the contrary, the SCILL before extraction exhibited lower uptakes for both gases compared to the SILP.

Taking into consideration the results of LN₂ porosimetry and CO₂, CO absorption of the SCILLs, as well as the amount of IL loaded onto the SILP (0.75 mL/g Figure 1a) and the aspects of oriented deposition of the functionalized IL molecules in the deposited IL layer (section 3.4), we have concluded about the possible configurations of the SCILL sample before and after the removal of the excess gold (Figure 9). In this regard, the amount of gold ions in the solution suffices to interact with the entire

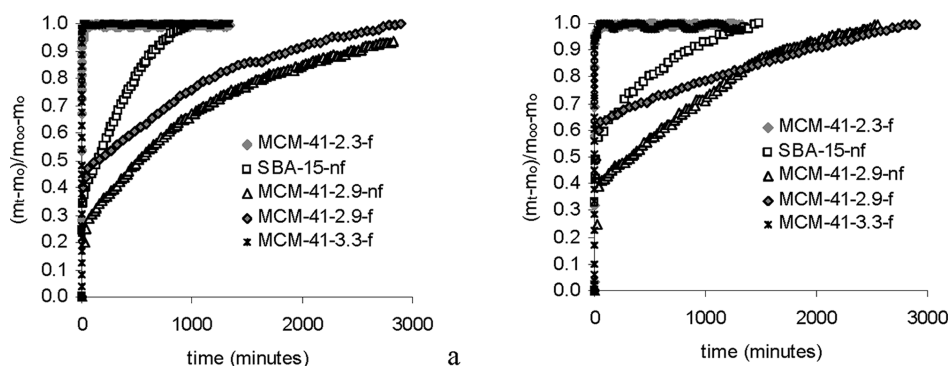


Figure 7. Transient curves of CO₂ adsorption of the several SILPs: (a) pressure step 0–250 mbar; (b) pressure step 250–500 mbar.

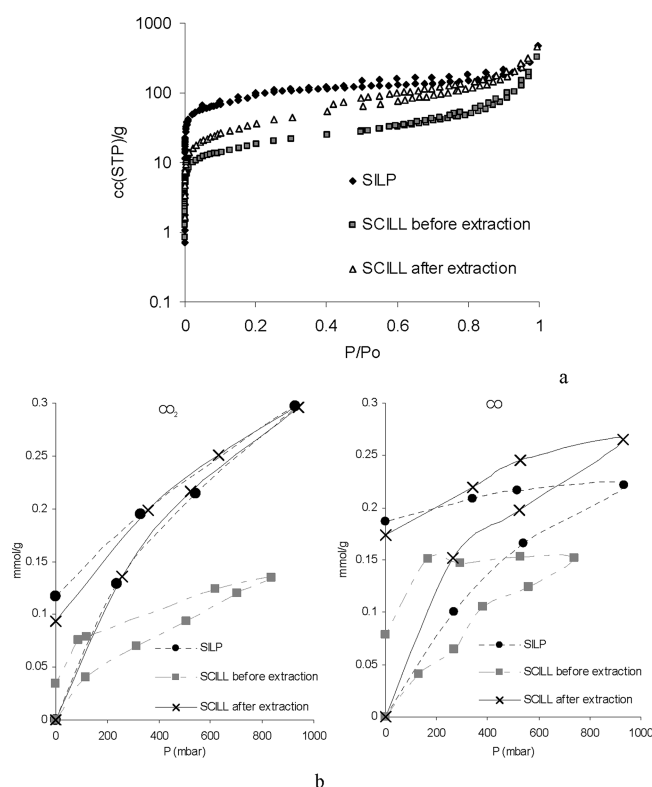


Figure 8. (a) LN₂ isotherms of the SILP and the respective SCILL before and after extraction of the excess gold. (b) CO₂ and CO absorption isotherms at 35 °C for the SILP and the respective SCILL before and after extraction.

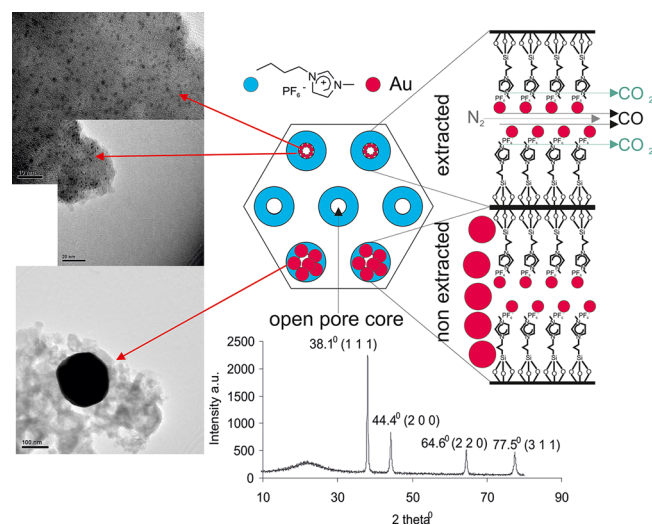


Figure 9. (left-up) TEM images of the ultrasmall (2 nm) AuNPs deposited inside the pores of the SCILL and the respective diffusion pathways of the several gases (right). The lower TEM image shows a large aggregation of AuNPs inhibiting the entrance of gases into the pore structure (schedule on the right-low). XRD analysis of the SCILL before extraction confirmed the existence of large particles.

population of PF₆ anions of the deposited into the pores IL monolayer. This interaction strongly stabilizes the gold nanoparticles (AuNPs) formed following the reduction with DMF, which exhibit a very small size of the order of 2 nm (TEM image, Figure 9) and decorate the top surface of the IL layer. However, there is also excess of gold cations in the solution that are loosely

anchored on the external surface of the SILP and form larger aggregates of AuNPs upon reduction with DMF. As indicated in Figure 10, the extracted SCILL possessed open diffusion paths for all the gases examined here. Carbon dioxide diffuses into the bulk of the pore through the PF₆ anions, and CO finds adequate space to interact with the AuNPs and form chemisorbed Au_m(CO)_x clusters.³⁹ Therefore, by subtracting the CO isotherm curve of the bare SILP at 308 K from the respective one of the extracted SCILL, it was possible in a simple way to deduce the % w/w active gold loading on the SILP, which was 1.25% w/w. Details on this isotherm subtraction procedure can be found in a previous publication of our group.⁴⁰

In the case of the nonextracted SCILL, aggregation of larger AuNPs on the external surface of the SILP in close proximity to the pore mouths partially inhibits the accessibility of gases to the IL and the internally deposited AuNPs. This is reflected in SCILL gas uptakes that are lower than those observed for the relevant SILP. In addition to TEM evidence, the existence of these large aggregates was also confirmed by Scherrer's analysis of the XRD data (Figure 9, size of 26 nm).

In this context, we can argue that the conditions of SCILL preparation, especially in what concerns the concentration of the metal salt and the mass ratio of the metal ions to the IL loaded on the SILP are decisive for the successful development of the catalytic system. Because the gold cations are anchored to the anions on the oriented IL layer, an approximate calculation of the optimum salt concentration can be done in relation to the population of the anions in the SILP, thus, avoiding implications of excessive loading and nanoparticle aggregation and the need for further post treatments to get rid of unwanted large and inactive particles.

3.6. Gas Separation Performance of SILPs. The gas separation performance was investigated in relation to the pore blocking extent, the IL molecules orientation, and the IL state (crystalline or liquid) in the deposited into the pores layer. Two gas molecules were studied, CO₂ and CO, one of which (CO₂) exhibits a confirmed specific interaction with the PF₆[−] anions. From the absorption isotherms presented in Figure 10a,c, it can be concluded that the CO₂/CO selectivity was significantly enhanced at lower temperatures. The reason is that at the temperature of 273 K, structuring in the IL films must have occurred, from the liquid phase at room temperature to the so-called IL crystalline phase (ILCs).^{41,42} This phase transformation renders the diffusion of the noninteracting CO molecule very slow and does not readily affect the absorption and diffusion properties of the specifically interacting CO₂. In a recent study,⁴³ the desired structural information of 1,3-didodecylimidazolium tetrafluoroborate [C12 C12Im][BF₄] spin-coated onto a clean Pt(111) single crystal was obtained from surface vibrational spectroscopy measurements at different temperatures (IRAS). Subtle spectral changes occurring around 344 K and below 318 K were attributed to the liquid–liquid crystalline (LC) phase transformation and the LC–solid transition, respectively. The short alkyl chains of the IL used in our work do not allow us to conclude that a solid transition took place. However, the significant improvement of the CO₂/CO separation performance of the SILPs upon lowering in temperature is an indication that an LC phase transformation must have occurred that reduces the interaction (solubility–diffusivity) of the IL phase with CO.

Another important observation from the absorption isotherms of Figure 10 is that complete pore blocking of the support is not sufficient for the development of a SILP system that will exhibit

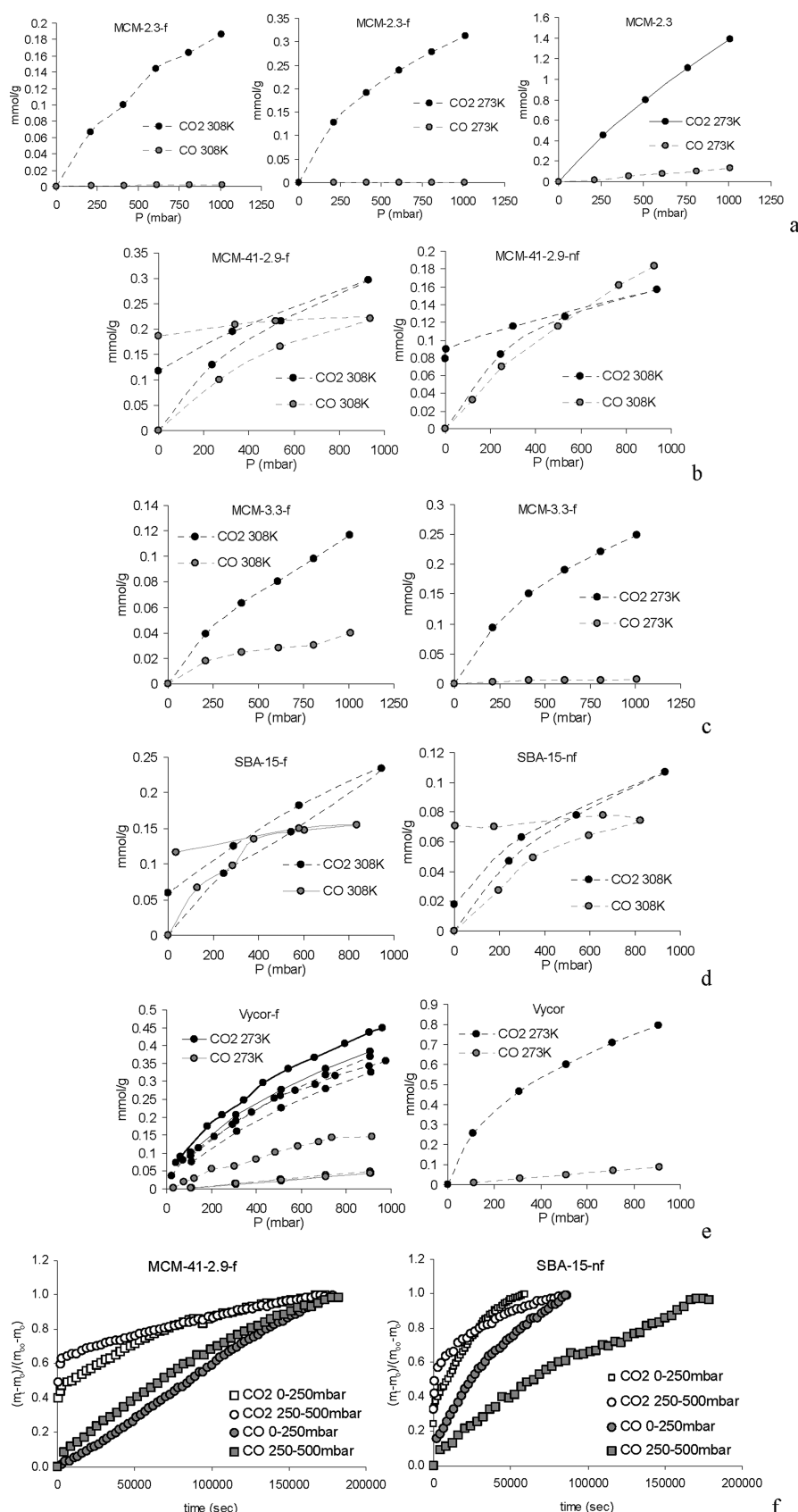


Figure 10. (a–e) Comparison between CO₂ and CO absorption capacities of the developed SILPs; (f) Transient curves of CO₂ and CO absorption for SILPs developed with a functionalized and a nonfunctionalized IL.

enhanced absorption selectivity for a gas interacting with the IL over a gas that does not present any specific interaction.

Complete pore blocking must be combined with ordering of the IL molecules in the deposited-into-the-pores layers in order to

achieve a high selectivity ratio. To get to this conclusion, one should compare the CO₂ and CO absorption isotherms of SBA-15nf (Figure 10d) and MCM-41–2.3-f (Figure 10a). Both samples exhibited complete pore blockage, as elucidated by SANS, but only the second sample gave signs of ordering and LC transformation in the respective XRD and MDSC analysis.

A feature that seems contradicting to the fact that CO does not present any specific interaction with the specific IL, is that the CO isotherm of the SILPs (Figure 10b,d) exhibited an enhanced hysteresis loop during desorption. Such behavior is characteristic of the strong binding of the adsorbate molecules to the absorbent phase. Despite this, the hysteresis observed in our case can be solely attributed to the very slow diffusion of CO. As can be seen from the transient curves of the first two points (pressure steps) of the absorption isotherms (Figure 10f), the diffusivity of CO was so slow that equilibrium was not reached, even after a period of more than 2 days. In this regard, during the subsequent lowering of the CO pressure, the sample may still absorb rather than desorb the CO to the gaseous phase.

As a general remark, complete pore blocking in combination with IL molecules ordering and LC transformation are required to develop a SILP system with enhanced selectivity of an interacting over a noninteracting with the IL, gas molecule. However, it could be possible to apply open pore SILPs as well as nonordered SILPs for a successful separation application in the case when the very different gas uptake rates are exploited (diffusion selectivity instead of equilibrium selectivity). Figure 11 evidence the proof of concept for this kind of separation.

Although the equilibrium CO₂/CO selectivity for samples MCM-41–2.9-f and SBA-15-nf is 1.3 and 1.5, respectively, there was a certain period (up to 1 h) during the transient state of each

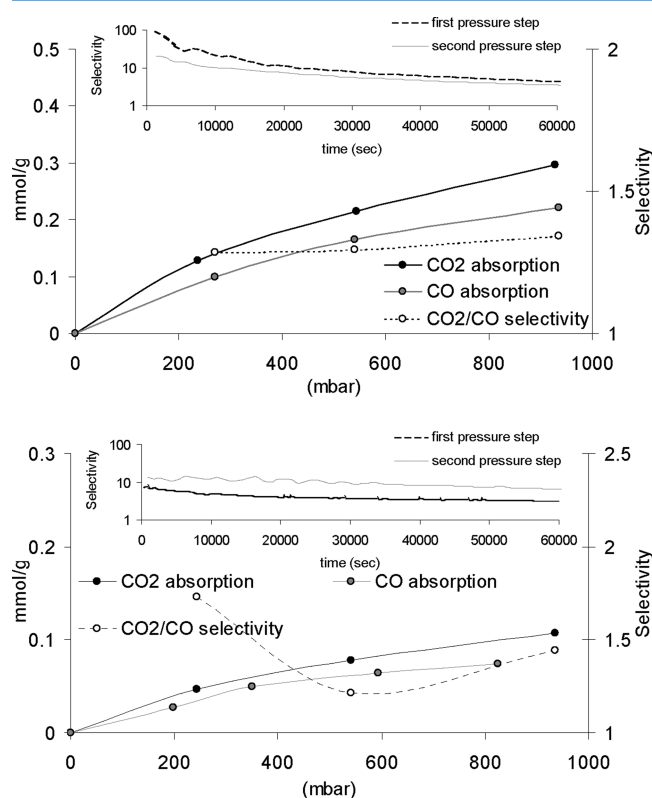


Figure 11. CO₂/CO separation performance based on equilibrium absorption and on diffusivity for samples MCM-41–2.9-f (top) and SBA-15-nf (bottom).

equilibrium point where the diffusion selectivity was well above 20. Of course when a gas separation process, for instance, a pressure swing adsorption (PSA) process, is based on the transient and not the equilibrium sorption performance of the applied material, someone must consider the lower amount of the selectively absorbed gas in combination with the separation factor, taking also into account all the process parameters such as the volumetric flux and composition of the gaseous stream, to end up with a decision on the feasibility of such a process.

3.7. Catalytic Performance. Thin films of ILs on a high surface area support were used to immobilize the AuNPs, and the low temperature oxidation of CO was applied as the process of choice so as to evaluate the catalytic activity of the developed SCILL. The MCM-41–2.9-f sample was involved as the most appropriate SILP support, which fulfilled the requirements of open porosity, to host the maximum possible amount of AuNPs and of IL monolayer and to avoid enhanced diffusion pathways of the gas reactants and products. Moreover, due to the preferred orientation of the IL molecules in the deposited layers of this specific SILP, it was possible to further exploit the interaction of the ILs anions with the gold cations. In this context, upon reduction of the Au³⁺ with DMF, the grafted IL molecules served as anchoring centers that ensured better dispersion and stability of the AuNPs during the catalytic reaction. In a recent work it was shown that BMIMPF₆ forms a protective layer surrounding the nanoparticles. This layer is composed of semiorganized imidazolium anionic supramolecular aggregates [(BMIM)_{x-n}(PF₆)_x]ⁿ⁻ located immediately adjacent to the nanoparticle surface and counter supramolecular cationic aggregates [(BMIM)_x(PF₆)_{x-n}]ⁿ⁺ providing charge balance.^{44,45} Electrostatic interaction provided by the intrinsic high charge of BMIM-PF₆ and steric hindrance due to the supramolecular anionic and cationic aggregates stabilize the AuNPs. Furthermore, the lack of charge balance in the area of the nanoparticle/(PF₆⁻) interface may trigger synergetic effects to the catalytic oxidation reaction. Another important interaction is this of the ILs anions with CO₂. Contrary to the usual practice of using ILs exhibiting enhanced solubility for the reactants (CO, O₂), our concept encompasses the use of an IL with enhanced affinity for the product of the oxidation reaction (CO₂).

Having ensured the maximization of the reactant concentration at the gold active centers, due to their inclusion into the narrow pores of the SILP, we have further examined the case of involving the IL as an “in situ trap” that abstracts the products from the region of their generation, a function that may considerably improve the yield of the reaction. It is notable that the incorporation of the AuNPs on to the IL surface into the pores of the SCILL catalyst resulted to the significant reduction of the difference between the uptake rates of CO₂ and CO (Figure 12a).

Due to the extreme narrowing of the pore openings in the SCILL systems, the gas diffusivities are very low and diffusion of the reactants and the products into and out of the pores, respectively, may be the rate-determining steps of a heterogeneous catalytic reaction. In this context, the use of a SCILL where the diffusivity of the reactants into the pores and on the nanoparticles surface is of the same order with the diffusivity of the products out of the pores may be very beneficial for the reaction.

The low temperature CO oxidation performance of the SCILL before and after extraction is presented in Figure 12b. The conversion (R%) of CO was calculated in two ways: (1) by taking into account the stoichiometry of the CO oxidation reaction

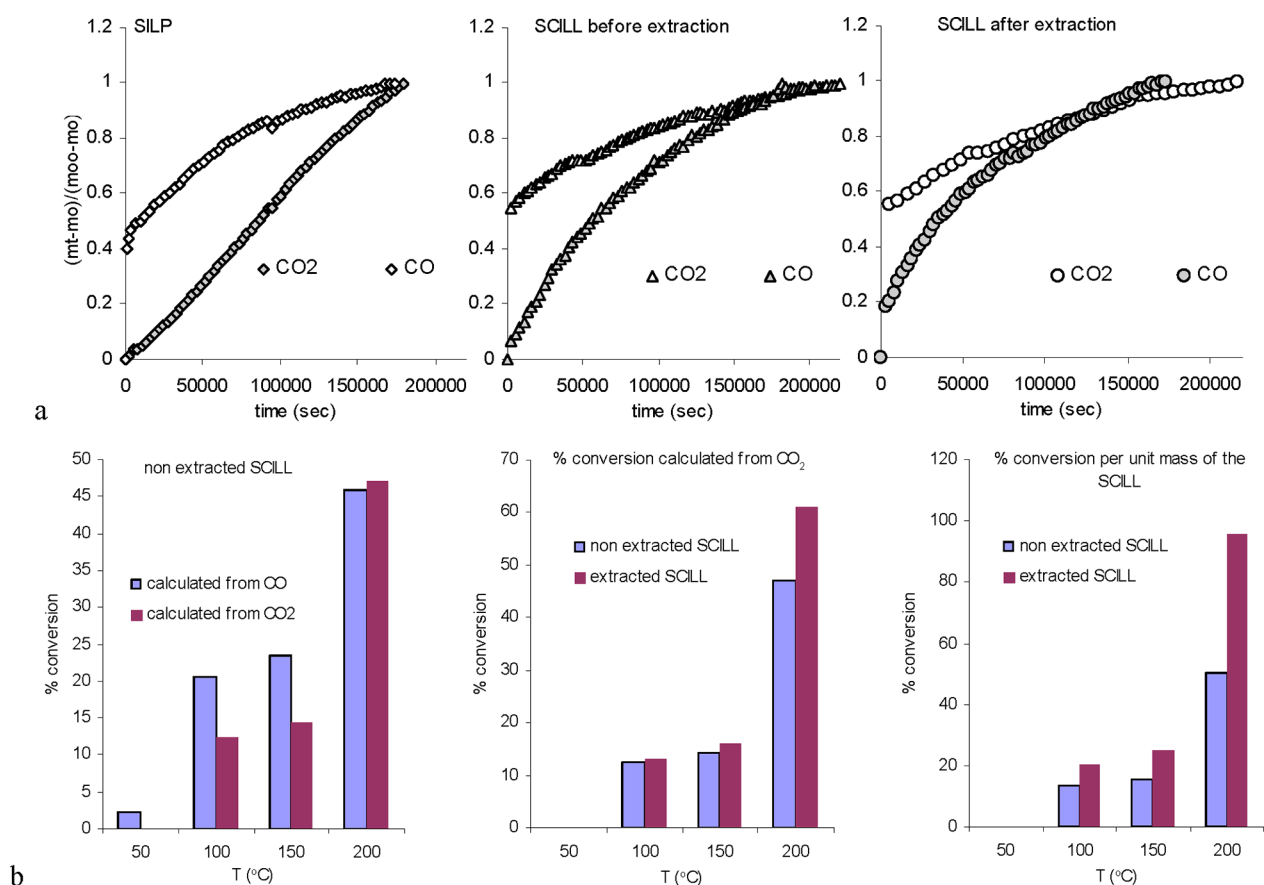


Figure 12. (a) CO diffusivity becomes faster after the deposition of the AuNPs on the IL layer inside the pore structure of the SILP, (b) CO oxidation efficiency of the developed SCILL, (left) comparison between the catalytic efficiency calculated as a function of the amount of CO converted and as a function of the amount of CO₂ produced, (center-right) comparison of the catalytic efficiency between the extracted and the nonextracted SCILL.

$$R\%_{\text{CO}} = 100 \times \frac{C_{\text{COinit}} - C_{\text{COfinal}}}{C_{\text{COinit}}}$$

where C_{COinit} and C_{COfinal} are the initial and final (upon completion of the reaction) concentration of carbon monoxide in the gas phase, or (2)

$$R\%_{\text{CO}} = 100 \times \frac{C_{\text{CO}_2\text{final}}}{C_{\text{COinit}}}$$

where $C_{\text{CO}_2\text{final}}$ is the final (upon completion of the reaction) concentration of carbon dioxide in the gas phase. Thus, the contribution of the absorption of CO₂ by the subjacent to the AuNPs IL layer is highlighted when presenting in comparison the conversion efficiencies derived by means of the above equations (Figure 12b, left). As the temperature rises, the differences between the conversion efficiencies are diminished. At the lower experimental temperature of 50 °C, and despite the concluded 2.2% conversion of CO, no CO₂ was detected in the gas phase. The reason behind is that due to the low reaction temperature, the CO conversion is moderate and conclusively the amount of CO₂ produced is small and probably suffices solely to occupy the active sites [PF₆[−]] of the deposited IL layer. As the temperature increases, higher amounts of produced CO₂ diffuse faster out of the pores through the mechanism described in section 3.4. The high concentration of CO₂ into the pores generates a significant driving force for diffusion, which is also favored by the moderate absorption capacity of the IL at the higher temperatures.

Figures 12b (center, right) demonstrate the higher activity of the extracted sample due to the higher accessibility of the reactants to the active AuNPs deposited inside the pores. Especially in Figure 12b (right), we present the conversion normalized per unit mass of the SCILL catalyst to make more evident the difference in the catalytic oxidation efficiency between the extracted and the nonextracted sample. Finally, it must be noted that the conversion of 13% at 100 °C observed for the extracted sample is of the highest reported in literature for gold nanoparticles loading at the level of 1% w/w.^{46,47}

4. CONCLUSIONS

IL-based composite systems (SILPs and SCILLs) with high potential in gas separation and heterogeneous catalysis were prepared by making use of highly porous materials as support and functionalized ILs as surface modifiers. The IL's cation was modified by adding appropriate functionalities that exhibit high affinity for the usually fully hydroxylated support surface, given that the main interactions of the IL with a gas are limited in the area around the IL's anion. In this respect, a preferred orientation of the IL molecules in the deposited IL layer was achieved, with the anion facing the core of the pore (IL/gas interface). The beneficial effect of the oriented IL deposition was proved by the high values of gas diffusivities, which were 2 orders of magnitude higher than these obtained for the SILPs modified with a nonfunctionalized IL. The density of the functionalized IL accommodation depended on the population of anchoring groups on the support surface. A dense accommodation in

combination with the oriented deposition generated liquid crystalline (LC) ordering as confirmed by XRD and MDSC analysis. Furthermore, LC ordering proved detrimental for the absorption capacity of a gas that does not interact with the IL, such as CO. In this respect, CO₂/CO absorption separation capacities of above 50 were obtained depending on the degree of LC ordering and the extent of pore blocking. Oriented deposition was also beneficial when a SILP was utilized as support to disperse catalytic nanoparticles and prepare a nanostructured IL-based catalyst (SCILL). In this case, the anions of the deposited IL interacted strongly with metal cations in the solution and stabilized derived-by-reduction gold nanoparticles with an average diameter of 2 nm. Among other unique properties of the resulting systems, catalytic experiments showed an additional functionality of the IL, acting as an “in situ trap” that preferentially abstracts the product (CO₂) from the reaction site, which can open new horizons for rational design of sorbents and catalysts based on the IL/porous systems developed.

■ ASSOCIATED CONTENT

■ Supporting Information

Synthesis of the nonfunctionalized IL [bmim][PF₆[−]] and extraction process of the gold agglomerates from the developed SCILL, as well as the preparation procedure of the samples for TEM, XRD, gas absorption, and porosimetry measurements, the accuracy of the gravimetric method of absorption, the preparation of the SCILL for the catalytic experiments, and the conditions of the MDSC analysis. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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